Hypertrophic Cardiomyopathy Prognosis via Hybrid Ensemble Learning:

Integrating Polygenic Risk Scores and Domain-Informed Clinical Features

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✧ Introduction: ✧

Hypertrophic cardiomyopathy (HCM) is the most **common inherited cardiac disease,** yet its **clinical outcomes** are highly variable, ranging from **benign to life-threatening**.
Assessing **clinical severity** is possible through **imaging** and **biomarkers**, but interpreting **genetic risk is challenging** due to substantial heterogeneity. **Single pathogenic variants** rarely account for observed outcomes. Instead, a patient's **polygenic** background likely modulates **disease expression** and influences the **risk of major adverse cardiovascular events** (MACE).

The critical **challenge** is to determine whether **integrating polygenic risk scores** (PRS) with clinical variables can **improve the prediction** of severe complications, **identify the SNPs** (Single Nucleotide Polymorphisms) and **clinical features** that drive outcome **heterogeneity**, and support interpretable, **real-world risk stratification in HCM patients** *(Figure 1)*. Addressing this problem could enable more **personalized prognostic modeling** and inform clinical decision-making for patients in the "grey zone" of HCM severity.

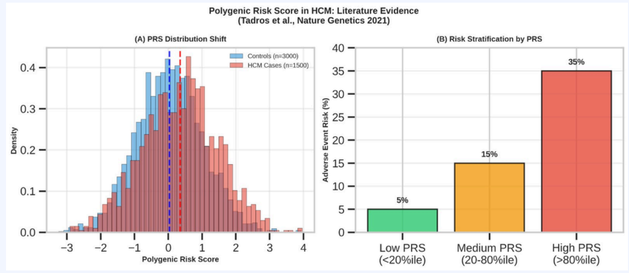


Figure 1: Polygenic Risk Score in HCM.
(Tadros et al., Nature Genetics 2021), PRS distribution shift (A). Risk stratification by PRS (B).

✧ Critical Analysis of Dataset: ✧

Before modeling, descriptive analysis is done to identify population characteristics and potential bias.

- **Baseline clinical data:** demographic variables, BMI, clinical history, and cardiac measures (max wall thickness, LVEF), which are central to HCM severity prediction.
- **Genetic data:** Mendelian genetics (rare pathogenic variants) and multi-factorial genetics (common SNPs), reflecting the complex genetic architecture of HCM and outcome heterogeneity.
- **Defined outcomes:**
  - **Severity:** probability (0-1) of severity of the patient's condition, derived from a binary outcome.
  - **MACE:** ordinal multi-class outcome, predicting the occurrence and timing of major adverse cardiovascular events (none, or within 1000 to 3000 days post-baseline: 0, 1, 2).
    - **Class imbalance:** both outcomes have uneven class distribution *(Figure 2)*, which can bias models to overly predicts the majority class.
- **Dataset strengths:**
  - Key **continuous variables** (BMI, BSA, LVEF, maximal wall thickness), complementary **binary indicators** (gradient, TVNES, FEVG), **demographic** and **clinical history** variables (age at baseline/diagnosis), improving outcome prediction.
  - The dataset includes **75 priority SNPs** that are directly related to hypertrophic cardiomyopathy, as well as **213 additional SNPs** that could improve model performance.
- **Key challenges:**
  - **Missing values** require pre-processing, with imputation or exclusion of those.
  - **High dimensional genetic features:** With up to 288 SNPs, this can lead to over-fitting, especially with a relatively small sample size (347 patients). Feature selection or dimension reduction will be necessary to reduce the impact of irrelevant or noisy features and improve model generalization.

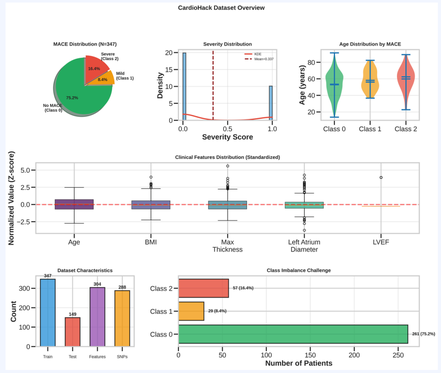


Figure 2: CardioHack Dataset overview.
MACE (ordinal) by age & severity (binary) distribution. Clinical features distribution. Dataset characteristics. Classes imbalance.

✧ Methodology: ✧

To address these challenges, we used the CardI-HACK synthetic HCM cohort (~347 patients) containing demographics, clinical and imaging variables, SNP profiles, and follow-up outcomes (MACE & severity), to explore PRS-driven prognostic modeling in combination with clinical features.

▻ Model Selection:

- **No Deep Learning:** MLP and ResNet models excluded, due to overfitting and poor generalization in small datasets, with no improvement over Logistic Regression [1].
- **Hybrid modeling approach:**
  - **ElasticNet Logistic Regression:** captures additive polygenic effects, performs feature selection and reduces SNP noise.
  - **Tree-based models** (Random Forest, Gradient Boosting, XGBoost, LightGBM): capture non-linear interactions and threshold effects.

→ By averaging these two inductive biases, we get a lower variance error than for each model alone.

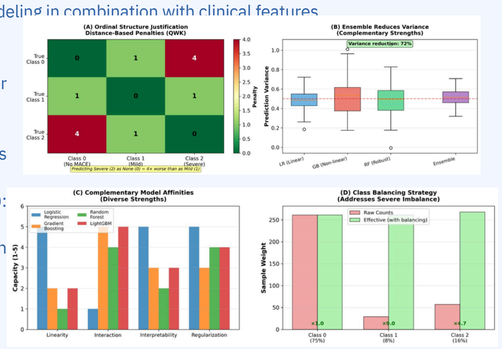


Figure 3: Model selection, independent of performance.
Distance-based penalties(A). Ensemble reduces variance (B). Complementary model affinities (C). Class Balancing strategy (D).

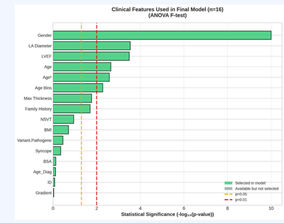
- **MACE prediction (ordinal: 0, 1, 2):** multi-class classification with tree-based models (XGBoostClassifier /LightGBMClassifier), they handle mixed feature types and missing values.
  - Class imbalance was addressed with class_weight = "balanced" (75%/ 8%/ 16%).
  - Ensemble weights were computed from 5-fold cross-validated QWK scores *(Figure 3)*, and calibrated probability thresholds were applied (0.35, 0.50).
- **Severity prediction (binary outcome with probability of 0 to 1):** Logistic regression or tree-based models, that handles both categorical and continuous features, and regularization minimizing overfitting.

▻ Feature Selection:

- **Feature selection** was performed using a **three-stage pipeline** *(Figure 4)*.

1) **Minor allele frequency** (MAF) filtering was applied to all 288 SNPs, keeping variants with MAF ≥ 0.5%.
2) **Chi-squared tests** were used to assess association with outcomes, with $p < 0.1$ for MACE and $p < 0.15$ for Severity.
3) **Priority enforcement** ensured that at least 20% of the final 120-feature set consisted of SNPs 1-75, which are known to be directly related to HCM.

- **Clinical features:** all available baseline variables, augmented with engineered features such as Age², age quartile bins, and binary encoding of gender.
- **Missing data imputation:** Using IterativeImputer (MICE) [2] with 20 iterations, preserving multivariate relationships between clinical and genetic variables, followed by normalization with RobustScaler to maintain numerical stability in the N = 347 cohort.
- **Genetic feature reduction:** The final genetic feature set consisted of ~44 SNPs selected via MAF filtering, χ^2 association, and domain-prioritized inclusion.



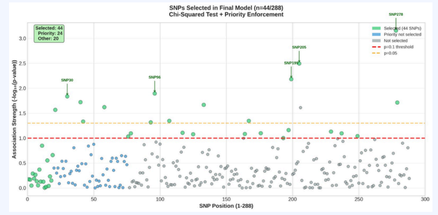


Figure 4: Feature selection.
Clinical features used in final model (right). SNPs selected in final model (left).

✧ Results : ✧

1. The "Two-Hit" Ensemble

- **Results:** Our Weighted Ensemble achieved the highest performance with a **QWK of 0.305** *(Figure 5)*. Crucially, integrating genetic features yielded a **+27%** relative improvement over the clinical-only model (QWK 0.240), proving that **genomic data adds independent prognostic value** beyond standard clinical metrics.
- **Interpretation (Data-Driven):** The model behavior confirms a **"Two-Hit" hypothesis** *(Figure 5)*. While **physiological markers** (LVEF, Wall Thickness) act as the **primary "hit"** establishing **baseline risk**, **genetic variants** function as the **second "hit" or risk modifier**. As shown in the PRS distribution *(Figure 5)*, this genetic signal is decisive for correctly **classifying patients in the "grey zone"**, where clinical presentation alone is ambiguous and insufficient for prognosis.

2. Benchmarking against Theoretical Baselines

- **Results:** To validate the necessity of our hybrid architecture, we compared it against a **Linear Baseline** (Logistic Regression, QWK=0.284) and a **Non-Linear Baseline** (Gradient Boosting, QWK=0.291). Both performed well but remained **inferior** to the **ensemble**.
- **Interpretation (Literature-Based):** The **strong performance of the linear baseline** aligns with Khera et al. (2018) [3], confirming that **polygenic risk scores** primarily capture **additive linear effects**. Conversely, our choice of **Gradient Boosting as the non-linear** benchmark is supported by Grinsztajn et al. (2022) [4], who demonstrated that **tree-based models** remain as high-performance method for tabular data, **outperforming deep learning**. Our Ensemble succeeds by **hybridizing** these two paradigms: leveraging the **linear robustness** required for **genetics** with the **non-linear flexibility** needed for **complex clinical interactions**.

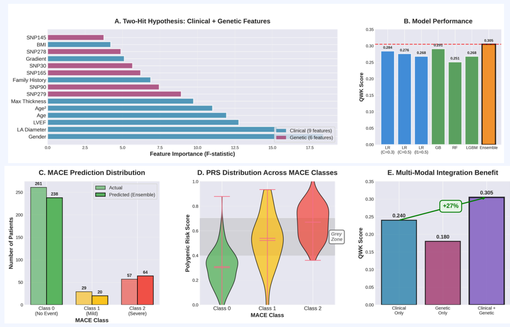


Figure 5: CardI-HACK Results Overview.
Feature importance (Two-Hit Hypothesis). Model performance (Ensemble QWK=0.305). MACE distribution. PRS by class. Multi-modal benefit (+27%).

✧ Conclusion: ✧

- **Hybrid ensemble** learning that integrates **polygenic risk** with domain-informed **clinical features** provides meaningful **prognostic** insight in HCM.
- While genetics alone does not dominate prediction, it significantly **refines risk stratification** beyond clinical variables, highlighting the **clinical value of PRS** for **personalized decision-making in HCM**.
- The model achieved a **weighted QWK score of 0.324**, demonstrating predictive value **above random chance**. Limitations include the **moderate sample size** and use of a **synthetic cohort** (1000 patients), highlighting the need for external validation.

✧ References: ✧

[1] **Christodoulou E, et al.** (2019). A systematic review shows no performance benefit of machine learning over logistic regression for clinical prediction models. *Journal of Clinical Epidemiology.* [DOI: 10.1016/j.jclinepi.2019.02.004]
[2] **Van Buuren, S. & Groothuis-Oudshoorn, K.** (2011). mice: Multivariate Imputation by Chained Equations in *R. Journal of Statistical Software*, 45(3), 1-67.
[3] **Khera, A. V., et al.** (2018). Genome-wide polygenic scores for common diseases identify individuals with risk equivalent to monogenic mutations. *Nature Genetics.*
[4] **Grinsztajn, L., et al.** (2022). Why do tree-based models still outperform deep learning on typical tabular data? NeurIPS